MUSIC GENRE CLASSIFICATION USING SUPERVISED MACHINE LEARNING

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PROBLEM STATEMENT

- This project uses machine learning to classify songs into genres based on audio features like tempo, loudness, and energy.
- The dataset is preprocessed, features are selected, and multiple models are trained.
- The performance of each model is compared using accuracy, and the best model is used to predict the genre of new songs

EXPLANATION OF THE PROJECT

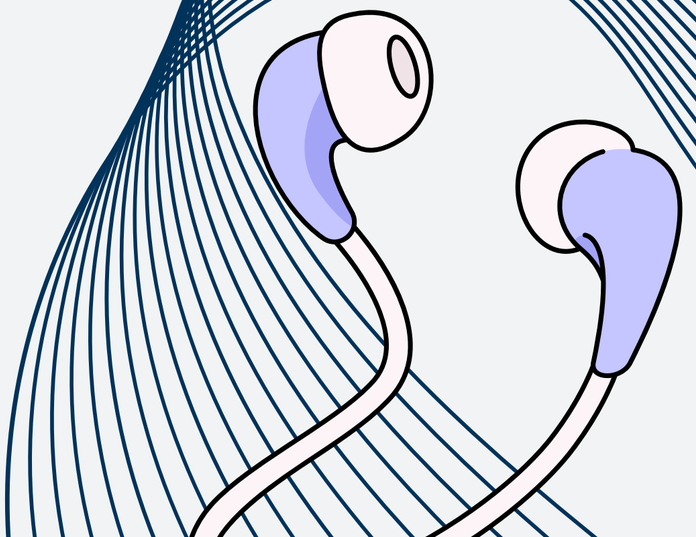
- Music datasets are large and difficult to classify manually.
- This project aims to build a machine learning model to automatically predict the genre of a song using audio features.
- Compare different algorithms, and improve accuracy using ensemble learning

DESCRIPTION OF THE ALGORITHMS USED:

This project uses three machine learning algorithms:

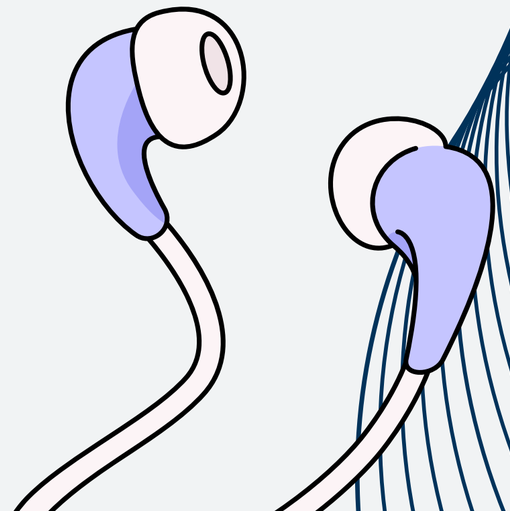
- Logistic Regression – a simple model used for classification
- K-Nearest Neighbors (KNN) – classifies data based on similarity to nearby points
- Support Vector Machine (SVM) – finds the best boundary to separate different genres

Voting Ensemble method is used to combine predictions from all models and improve overall performance

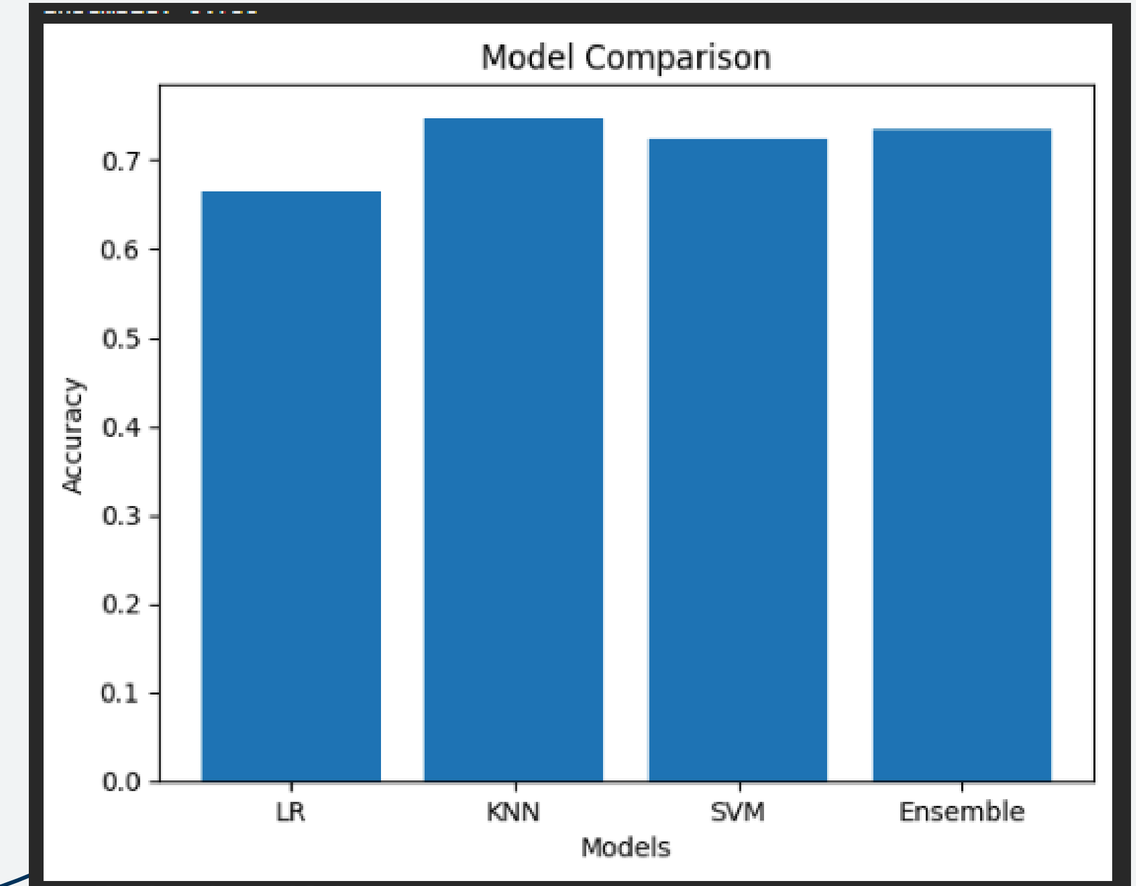


ACCURACY OF THE MODEL

- Logistic Regression: 0.665
- KNN: 0.748
- SVM: 0.725
- Ensemble: 0.735



FIGURES/PLOTS SHOWING RESULT:

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Prediction (Ensemble):
pop
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OUTCOME & CONCLUSION OF THE PROJECT

- The project successfully classifies music genres using machine learning techniques.
- Among the models, KNN achieved the best performance.
- While ensemble learning also improved overall results. The model can be further enhanced by using more features and a larger dataset.

THANK YOU